Verbindungen über Netznamen. Lesbare Zeichnung: hardware/*/testaufbau/*.svg

VBAT_SW
GND
J3
Batterie (nach S1)

+5V
D1
VBAT_SW
1N5817

R1
+5V
BATT
100k

R2
BATT
GND
47k

C1
BATT
GND
100n

R3
+3V3
A3V3
10

C2
A3V3
GND
100u/10V

C3
A3V3
GND
100n

R4
A3V3
VREF
10k

R5
VREF
GND
10k

C4
VREF
GND
10u

C5
A3V3
GND
100n

J1
1 VREF
2 COIL
3 GND
Spule (Anfang, Ende, Schirm)

R6
COIL
VREF
22k

R7
COIL
INP
1k

C6
INP
VREF
100p C0G

U1A
INP
INN_A
OUT_A
MCP6002-I/P

U1B
OUT_A
INN_B
OUT_B
MCP6002-I/P

A3V3
U1C
V+
V−
GND
MCP6002-I/P

R8
OUT_A
INN_A
47k

C7
OUT_A
INN_A
100p C0G

R9
INN_A
RG_A
10k

C8
RG_A
VREF
47n Folie

R11
OUT_B
INN_B
390k

C10
OUT_B
INN_B
22p C0G

R12
INN_B
RG_B
10k

C11
RG_B
VREF
47n Folie

R10
OUT_A
ADC_LO
1k

C9
ADC_LO
GND
1n

R13
OUT_B
ADC_HI
1k

C12
ADC_HI
GND
1n

R14
PIEZO_DRV
PIEZO
100

PIEZO
GND
J5
Piezo

J2
1 GND
2 +3V3
3 SCL
4 SDA
OLED (GND VCC SCL SDA)

J4
1 BTN_REF
2 BTN_MOD
3 GND
Tasten (REF MODUS GND)

J6
1 RES1
2 RES2
3 A3V3
4 VREF
5 GND
Reserve 2. Spule

A1
+5V 1 — 5V
GND 2 — GND
+3V3 3 — 3V3
PIEZO_DRV 4 — GPIO13
BTN_MOD 5 — GPIO12
BTN_REF 6 — GPIO11
SCL 7 — GPIO10
SDA 8 — GPIO9
9 — GPIO8
10 — TX/GPIO43
11 — RX/GPIO44
12 — GPIO1 — ADC_LO
13 — GPIO2 — ADC_HI
14 — GPIO3
15 — GPIO4 — BATT
16 — GPIO5 — RES1
17 — GPIO6 — RES2
18 — GPIO7
ESP32-S3-SuperMini

PWR_FLAG GND
PWR_FLAG VBAT_SW
PWR_FLAG +5V
PWR_FLAG +3V3
PWR_FLAG A3V3

Entwurf vor dem Mini-Test. Modulbelegung vor der Bestellung am echten Modul prüfen.

Kabelsuchgerät (Variante Einfach)

Sheet: /
File: handgeraet.kicad_sch

Title: Kabelsuchgerät - Handgerät

Size: A3	Date: 2026-10-13	**Rev: Entwurf 1**
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